Jiamin Pan

Ph.D. Candidate in Mathematics

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Education

01/2022 – 12/2026 (expected) Ph.D. in Mathematics, Georgia State University
Advisor: Florian Enescu; GPA: 4.22/4.30
Dissertation (Tentative): *The Behavior of Frobenius Complexities of Local Rings Under Standard Algebraic Operations*
(07/06/2026 – 07/18/2026) **SLMath Graduate Summer School: Singularities in Commutative Algebra Through Cohomological Methods**

08/2019 – 12/2021 M.S., Mathematics, Georgia State University
Concentration: *Statistics*; GPA: 4.18/4.30

09/2015 – 06/2019 B.Eng., Software Engineering, Huazhong University of Science & Technology
(08/2018 – 06/2019) **Graduate Preparation Program in Engineering (GPP-E)**
Major: *Computer Science*; University of California, Riverside, CA; GPA: 3.79/4.00

Research Interests

Commutative algebra, combinatorial matrix theory, and computational methods in algebra.

Publications and Preprints

Published / Accepted

1 Florian Enescu, Jiamin Pan, *On the finiteness of the twisted complexity for a graded ring in prime characteristic*, Bulletin Mathématique de la Société des Sciences Mathématiques de Roumanie, Tome 69 (117), No. 3, 2026, pp. 289-300.

2 Florian Enescu, Priyank Kalla, and Jiamin Pan, *Computing Boolean functions from polynomial ideals with prescribed zeros*, Journal of Algebra and Its Applications, 2025.

3 Marina Arav, Frank J. Hall, Hein van der Holst, Zhongshan Li, Zixuan Li, Jiamin Pan, Hanfei Xu, Yiran Xu, Zheng Yang, *Orthogonal similarity and transversal intersection of manifolds*, Communications in Optimization Theory, Vol. 2025 (2025), Article ID 27, pp. 1-21.

4 Marina Arav, Frank J. Hall, Hein van der Holst, Zhongshan Li, Aram Mathivanan, Jiamin Pan, Hanfei Xu, Zheng Yang, *Advances on similarity via transversal intersection of manifolds*, Linear Algebra and its Applications, Volume 721 (2025), 102-121.

Submitted and Preprints

5 Marina Arav, Frank Hall, Alfred Hart, Hendricus van der Holst, Zhongshan Li, Zixuan Li, Jian Liu, Jiamin Pan, Joonwon Seo, Li Wang, Hanfei Xu, Yiran Xu, Yue Zhao, Zheng Yang, *The rank-preserving transversality property*, Submitted to Linear and Multilinear Algebra, 2026.

6 Marina Arav, Frank J. Hall, Alfred C. Hart, Hein van der Holst, Zhongshan Li, Jian Liu, Jiamin Pan, Joonwon Seo, Li Wang, Hanfei Xu, Yiran Xu, Yue Zhao, Zheng Yang, *Patterns that require the rank-preserving transversality property*, in preparation.

7 Jiamin Pan, *The Frobenius complexity of rings of essentially finite type*, in preparation.

Talks and Presentations

09–10/2026 *A finiteness criterion for Frobenius complexity, I–III*, Departmental Algebra Seminar, Georgia State University.

04/2026 *Uniformly F-compatible ideals, I & II*, Departmental Algebra Seminar, Georgia State University.

03/2026 *Properties of Uniformly F-Compatible Ideals*, 105th Annual MAA-SE Section Meeting, University of North Alabama.

04/2025 *Computing Boolean Functions from Polynomial Ideals with Prescribed Zeros*, Commutative Algebra in The South, Georgia State University.

04/2025 *Matrices with the Rank-Preserving Transversality Property*, Departmental End-of-the-Year Celebration, Georgia State University.

Awards and Honors

2025 V.V. Lavroff Graduate Award ($700)

2024 Wiley M. Suttles Math Award ($520)

2023 Best Graduate Teaching Assistant Award, College of Arts & Sciences

2023 Wiley M. Suttles Math Award ($500)

Teaching Experience

01/2026 Substitute Lecturer, Georgia State University
- MATH 2212: Calculus of One Variable II (Spring 2026, three weeks of lectures)
- Wrote Test 1 (Two Versions) and the accompanying grading rubrics

06/2025 – 07/2025 Course Materials Development, Georgia State University
- Revised the recitation worksheets for MATH 2211 & 2212 - Calculus I and II: corrected errors, ordered exercises by difficulty, and added problems for topics not yet covered.
- All adopted for the subsequent semesters.

01/2021 – 12/2022 Instructor of Record, Georgia State University
05/2023
06/2024 – 07/2024
- MATH 2211: Calculus of One Variable I (Summer 2024)
- MATH 1111: College Algebra (Fall 2021, Spring 2022, Fall 2022, Summer 2023)
- MATH 0999: Support for College Algebra (Spring 2022)
- MATH 1001: Quantitative Reasoning (Spring 2021)

08/2023 – present Recitation Instructor, Georgia State University
- MATH 2212: Calculus of One Variable II (Fall 2023, Spring 2024, Spring 2025, Fall 2025, Spring 2026, Fall 2026)
- MATH 2211: Calculus of One Variable I (Fall 2024)

08/2019 – 12/2020 Graduate Laboratory Assistant (Tutoring), Georgia State University
06/2026 – 07/2026
- MATH 1401: Elementary Statistics (2019–2020)
- MATH 1113: Precalculus (Summer 2026)

Professional Service

09/2025 Application Reviewer, Math Path Fall Assistantship

02/2025 Application Reviewer, D-Map Summer Program

05/2022 Application Reviewer, Math Path Summer Program

Other Professional Experience

06/2025 – 08/2025 AI Trainer – Mathematics Expert
Handshake Model Validation Expert (MOVE) Fellowship (Now: Handshake AI)

- Wrote and researched specialized mathematics prompts to evaluate the reasoning of large language models.
- Reviewed model outputs for mathematical correctness, rigor, and clarity, and gave expert feedback to improve them.

Computational Skills

Computer algebra	Macaulay2, Singular
Programming	Python, R, C, C++, Java, MATLAB
AI tools	LLM assistants for literature exploration and drafting code, with independent verification of all mathematical claims and computational output
Typesetting	LaTeX, Markdown